

RBS Project - R

Telco Customer Churn: the Story Told by the Visuals

Part 1: A visual story of the analysis (13 visuals)

Part 2: The project questions, answers, explanations and visuals

Data: IBM Telco Customer Churn dataset (7,043 customers, California). 26.5% of customers churned (1,869 customers). All numbers and plots in this document come from the R Markdown file *Telco_Churn_EDA.Rmd*.

Part 1 - The Story of the Visuals

This part walks through every visual of the R code in the order the analysis was built. For each one we say **what the chart shows** and **what it means for understanding churn**. Read together, the visuals tell one story: *who leaves, why they leave, and what keeps customers loyal*.

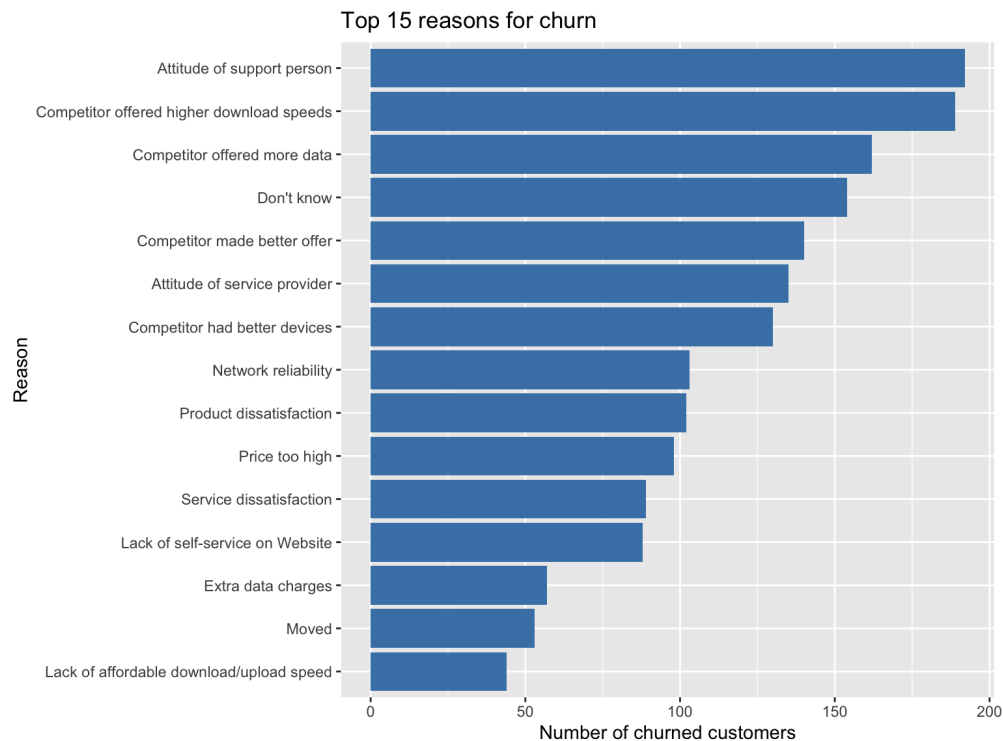
The story in one paragraph

About one in four customers leaves. When asked why, they mostly point to **competitors** and to the **quality of staff and support**. The customers most at risk are those on month-to-month contracts, paying by electronic check, using fiber optic internet, living alone, and without protection services such as Online Security or Tech Support. Customers with dependents, longer contracts and more protective services almost never leave. Price level and location turn out to be much weaker explanations than the type of product and the type of relationship the customer has with the company.

Chapter 1 - Why do customers leave?

We start with the customers who left and listen to what they said.

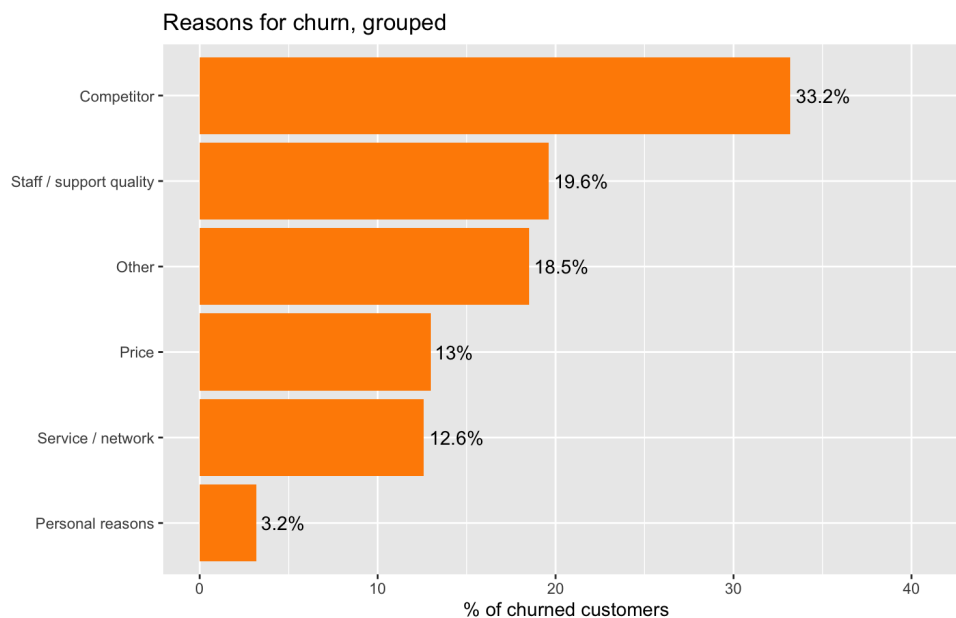
Visual 1 - The 15 most common churn reasons



What the chart shows. Each bar is one reason given by customers who churned. The longest bars are 'Attitude of support person' (192 customers), 'Competitor offered higher download speeds' (189), 'Competitor offered more data' (162) and 'Don't know' (154).

What it means. Four of the top seven reasons start with the word 'Competitor' and two are about staff attitude. Customers do not mainly say 'it is too expensive' (Price too high is only 98 customers). They leave because someone else offers a better package, or because they were treated badly. With 20 different reasons the chart is hard to summarise, so the next visual groups them.

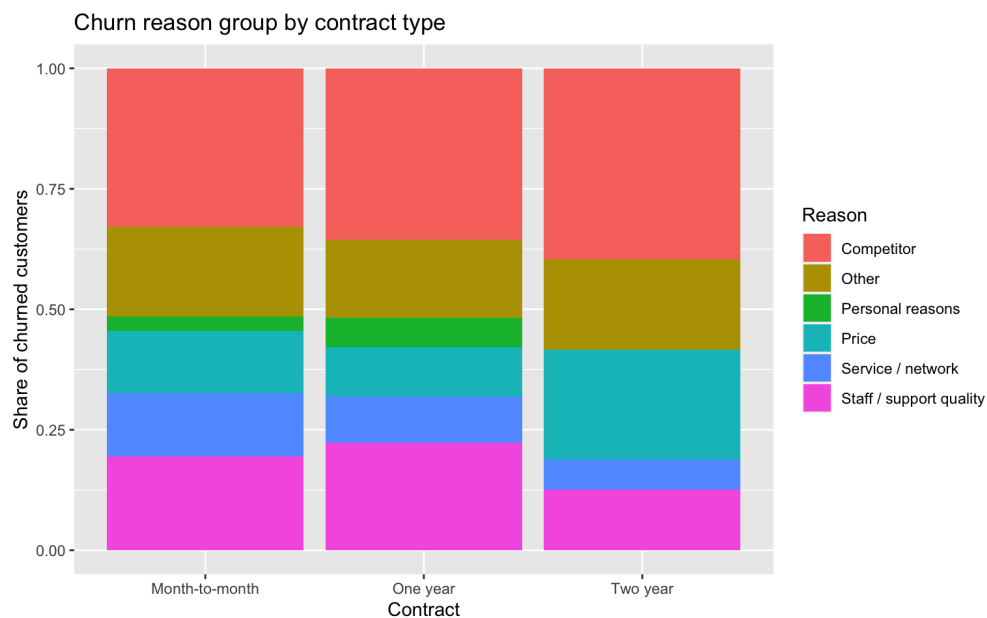
Visual 2 - Churn reasons grouped into six categories



What the chart shows. The 20 reasons are grouped by keyword into six categories and shown as a share of all churned customers: Competitor 33.2%, Staff / support quality 19.6%, Other 18.5%, Price 13.0%, Service / network 12.6% and Personal reasons 3.2%.

What it means. One third of churn is about competitors, and one fifth about the way staff and support treat customers. Together that is more than half, and both are things the company can influence. Only 3.2% leave for personal reasons such as moving, which the company cannot control. This is the first sign that churn is largely *preventable*.

Visual 3 - Churn reasons by contract type



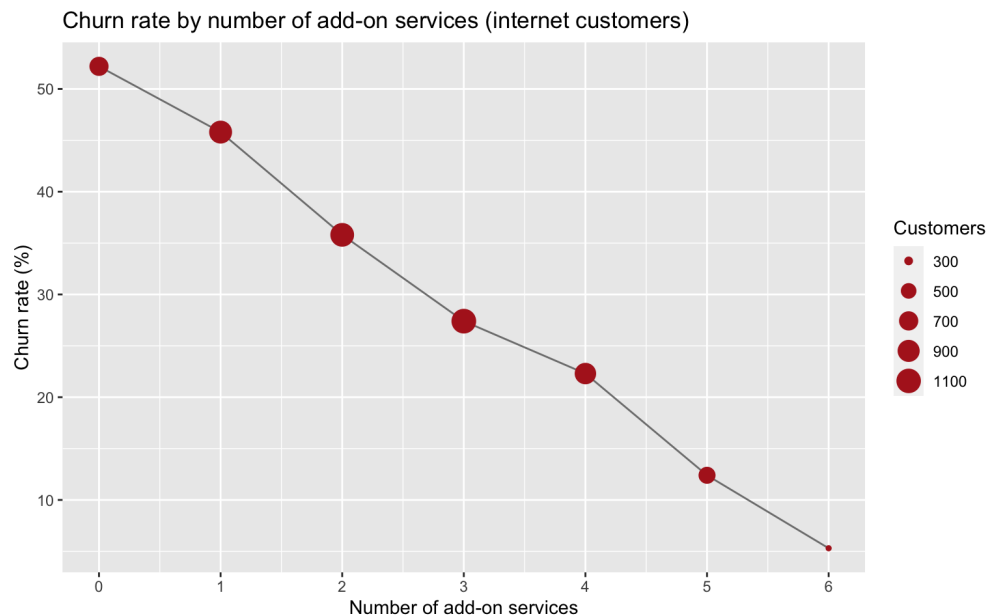
What the chart shows. Each bar is a contract type (month-to-month, one year, two year) split into the six reason groups, showing the share of churners who gave each reason. The competitor share is the biggest block in all three contracts (about one third to 40%).

What it means. The reason mix is broadly similar across contracts, so contracts change *how many* customers leave far more than *why* they leave. Caution: only 166 one-year and 48 two-year customers churned, so their bars are based on small groups. The price block looks larger for two-year customers, but with 48 people this is not reliable.

Chapter 2 - What keeps customers loyal?

Next we look at the customers who have add-on services.

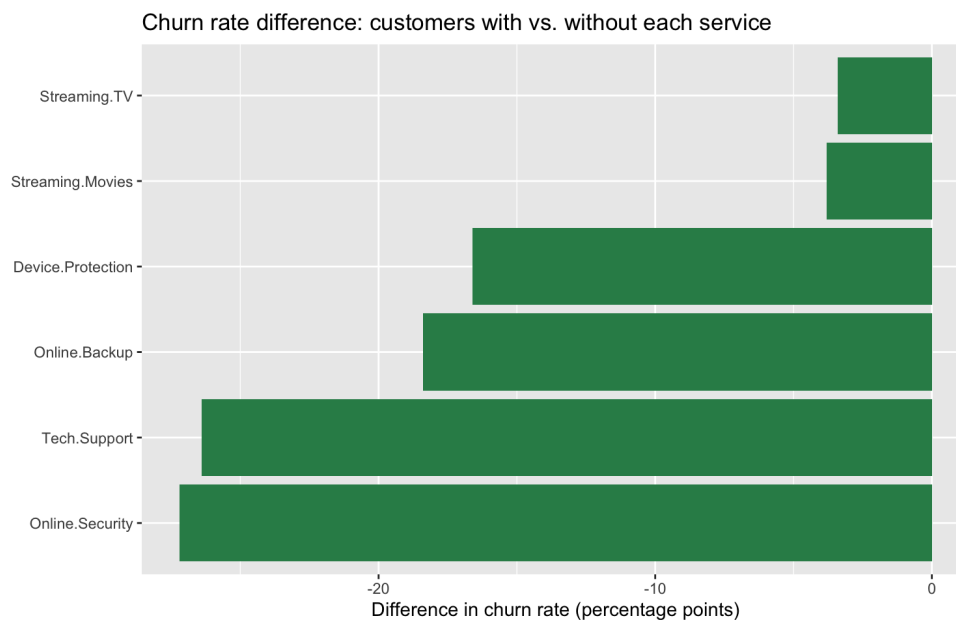
Visual 4 - Churn rate by number of add-on services



What the chart shows. Each point is a group of internet customers with 0 to 6 add-on services. The line falls almost straight: 52.2% churn with 0 services, 45.8% with 1, 35.8% with 2, 27.4% with 3, 22.3% with 4, 12.4% with 5 and only 5.3% with all 6. Bigger points mean more customers.

What it means. The more services a customer has, the less likely they are to leave. The fall is steady (the trend test is highly significant), so each extra service seems to tie the customer a little more to the company. This could also mean that already-happy customers buy more, so the chart shows a link, not proof of cause.

Visual 5 - Which services matter most?



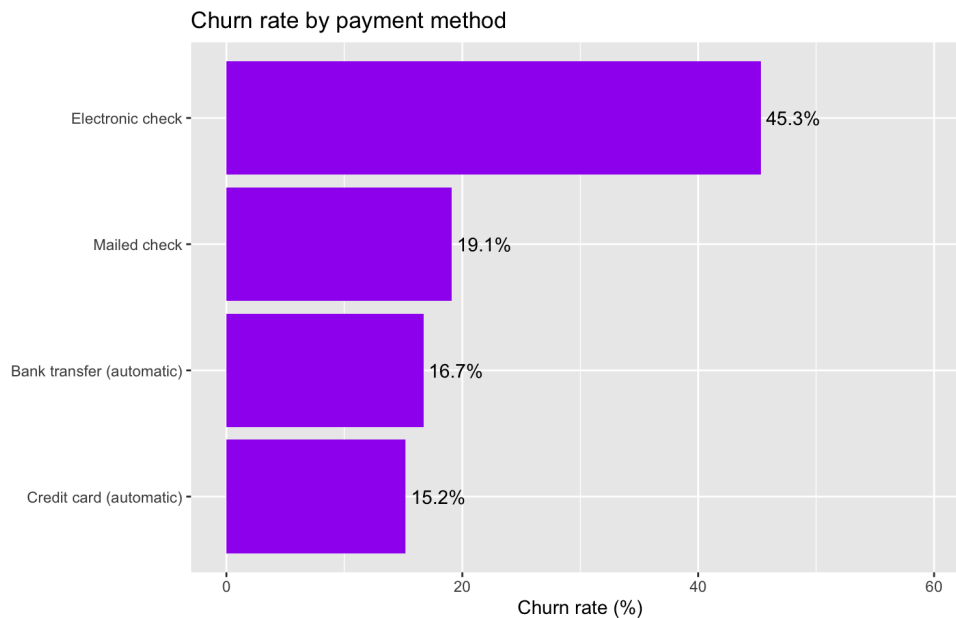
What the chart shows. For each service, the bar is the churn rate of customers *with* the service minus the churn rate of customers *without* it. Online Security (-27 points) and Tech Support (-26 points) have the biggest effect, then Online Backup (-18) and Device Protection (-17). Streaming TV (-3) and Streaming Movies (-4) hardly matter.

What it means. It is not the number of services but their **type**. Services that protect or help the customer build loyalty; entertainment extras do not. A practical idea for the company: offer security and support to customers who do not yet have them, especially new customers.

Chapter 3 - How customers pay

Billing looks boring, but it separates loyal and risky customers sharply.

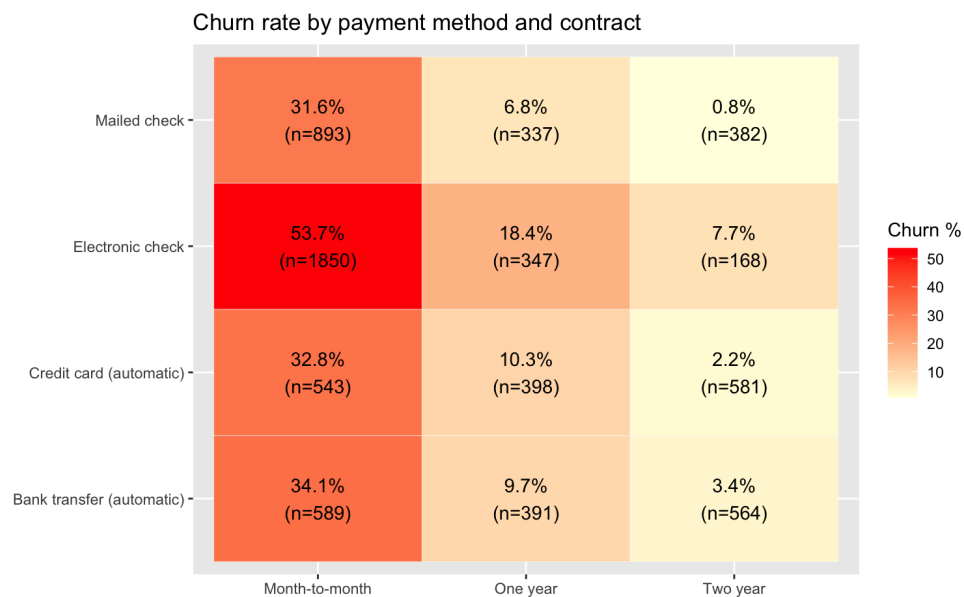
Visual 6 - Churn rate by payment method



What the chart shows. Electronic check customers churn at 45.3%. The other three methods are far lower: mailed check 19.1%, bank transfer (automatic) 16.7%, credit card (automatic) 15.2%.

What it means. Electronic check is a strong warning sign: churn is almost three times higher than for automatic payments. The chi-square test confirms that the difference is not due to chance. One explanation is that automatic payment is a habit that is harder to stop, while electronic check customers actively pay each month and can leave more easily.

Visual 7 - Is it really the payment method? Payment x contract



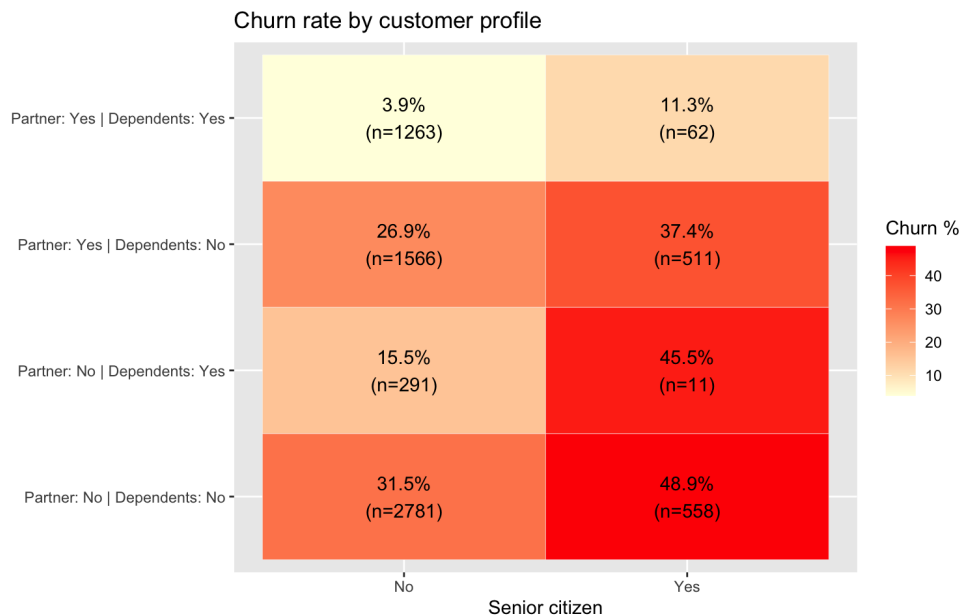
What the chart shows. A heatmap where colour shows churn rate and each cell shows the churn percentage and number of customers. The darkest cell is electronic check with a month-to-month contract: 53.7% (1,850 customers). The lightest cells are two-year contracts with mailed check (0.8%) or credit card (2.2%).

What it means. Contract matters enormously (churn falls from about 30-54% to 1-8% as the contract gets longer), but electronic check is the highest row *inside every contract type* (53.7%, 18.4%, 7.7%). So electronic check is not just a stand-in for month-to-month contracts; it carries extra risk of its own.

Chapter 4 - Who is the customer?

Now the customer's personal situation.

Visual 8 - Churn by customer profile (senior, partner, dependents)



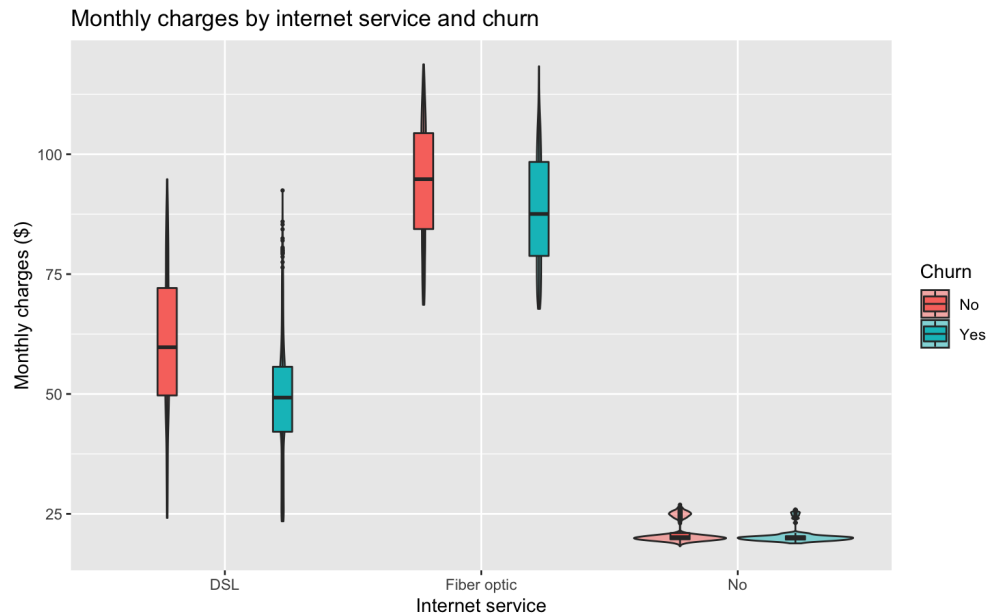
What the chart shows. Rows combine having a partner and dependents; columns are senior citizen or not. The highest churn is seniors with no partner and no dependents (48.9%, 558 customers). The lowest is non-seniors with both partner and dependents (3.9%, 1,263 customers).

What it means. Living alone raises risk, and having a family lowers it sharply, especially dependents (6.5% churn with dependents versus 32.6% without). Seniors churn more in almost every row. The cell 'seniors with dependents but no partner' has only 11 customers, so we ignore it. Gender, in contrast, is unrelated to churn ($p = 0.49$).

Chapter 5 - Price and product

Do customers leave because they pay more?

Visual 9 - Monthly charges by internet service and churn

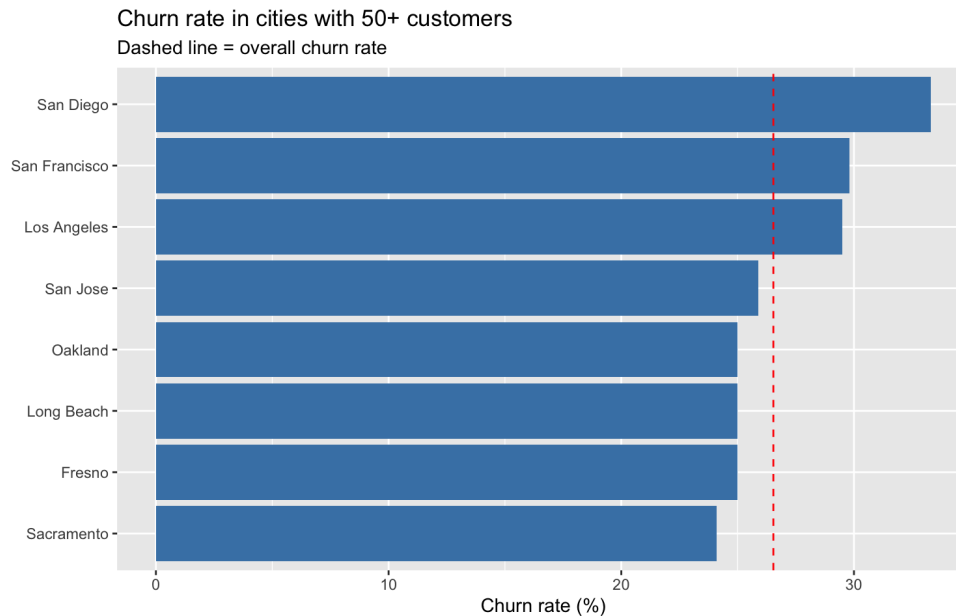


What the chart shows. Violin plots with boxplots inside, for each internet type, split into churned (blue-green) and stayed (red). Fiber optic customers pay about \$90-95 a month, DSL customers about \$50-60, and customers without internet about \$20. Inside DSL and fiber optic, the churned boxes sit *lower* than the staying boxes.

What it means. Churners pay more overall (\$74 vs. \$61 a month) only because they are more often fiber optic customers, who pay much more and churn much more (41.9% churn for fiber, 19.0% for DSL, 7.4% for no internet). Within the same product, churners actually pay slightly less. So price level alone does not drive churn; the fiber optic product does. (The violin shapes are very thin because the data has few repeated price points, so the boxplots are the easier part to read.)

Chapter 6 - Where do they live?

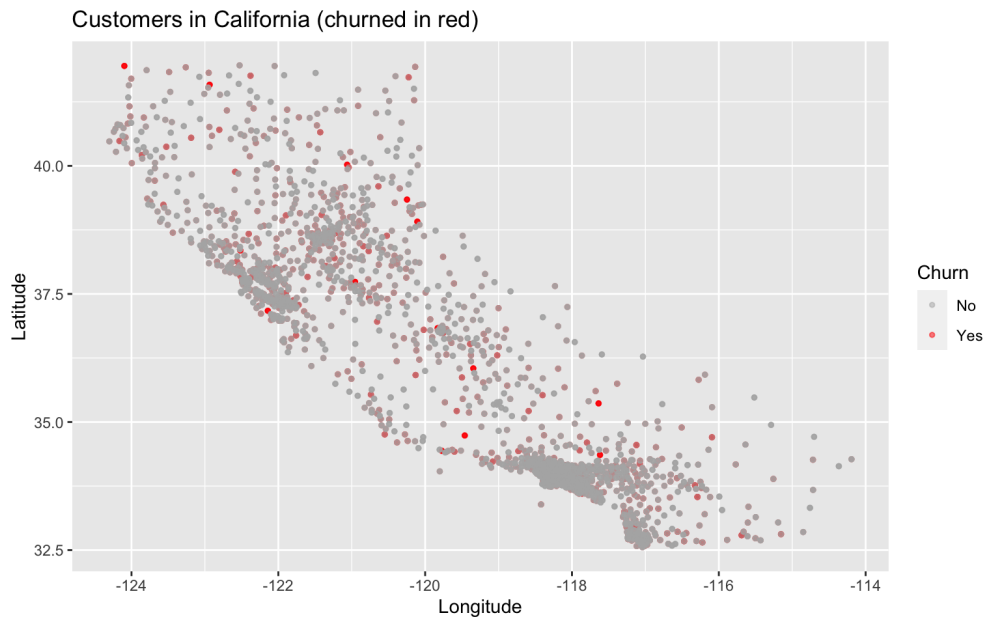
Visual 10 - Churn rate in the larger cities



What the chart shows. Bars show churn rate for the eight cities with at least 50 customers; the red dashed line is the overall churn rate (26.5%). San Diego is highest (33.3%), Sacramento lowest (24.1%).

What it means. The cities differ by a few points, but the chi-square test gives $p = 0.73$, so these differences are no bigger than we would expect from random variation. Location is not a main driver of churn.

Visual 11 - Map of customers in California

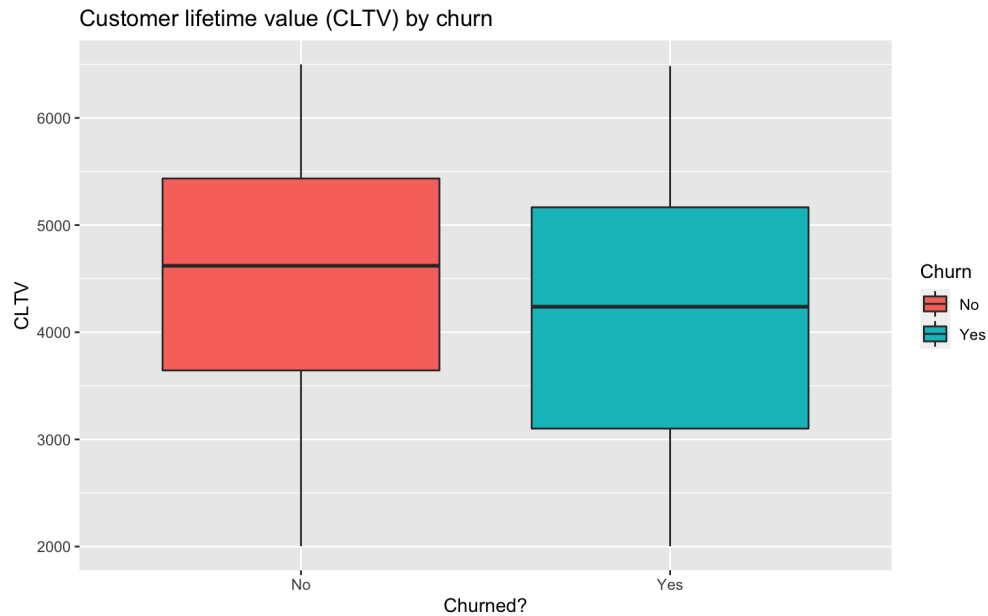


What the chart shows. Every dot is a customer, placed by latitude and longitude; red dots churned. Customers are concentrated around Los Angeles, San Diego and the Bay Area.

What it means. The red dots appear in every region and no area is clearly red. In the crowded areas grey dots are drawn over many red ones, so the map is only a rough check, but together with the city test it supports the conclusion that churn is spread evenly across California.

Chapter 7 - What does churn cost?

Visual 12 - Customer lifetime value (CLTV) by churn

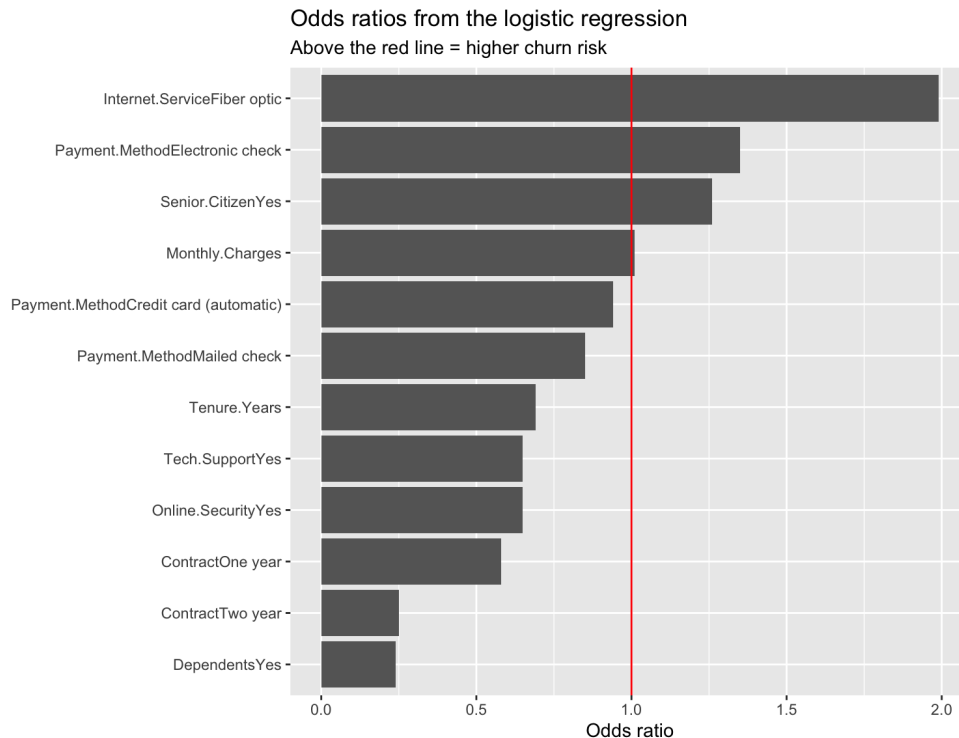


What the chart shows. Boxplots of CLTV for customers who stayed and customers who left. The median CLTV is about \$4,620 for those who stayed and \$4,240 for those who left, and the whole box of churners sits lower.

What it means. Customers who leave are, on average, less valuable over their lifetime (mean \$4,149 vs. \$4,491), and the difference is statistically significant. But the boxes overlap a lot, so CLTV alone does not identify who will leave. In money terms, churners represent 30.5% of monthly revenue (\$139,131 per month) while being 26.5% of customers, and 87% of that lost revenue comes from month-to-month contracts.

Chapter 8 - Everything together

Visual 13 - Odds ratios from the logistic regression



What the chart shows. Each bar is the odds ratio of one factor after controlling for all the others. A bar to the right of the red line (above 1) raises churn risk; to the left (below 1) lowers it. Fiber optic (1.99), electronic check (1.35) and senior citizen (1.26) raise risk. Dependents (0.24), two-year contract (0.25), one-year contract (0.58), Online Security (0.65), Tech Support (0.65) and each extra year of tenure (0.69) lower it.

What it means. This confirms the earlier charts in one picture, even when factors are considered together. The strongest protections are having dependents and a long contract; the strongest risks are fiber optic internet and electronic check payment. Monthly charges barely matter once the others are included (odds ratio about 1.01 per dollar). On 30% of customers that the model had not seen, it classified 77% correctly, versus 69% by always guessing 'No'.

Part 2 - Project Questions and Answers

The assignment asked for an independent exploratory analysis of the Telco churn data, with our own analytical questions, appropriate statistics, visualizations, and a summary of findings. Below are the eight questions we chose, each with a short answer, the explanation and the key visuals.

Overview of the statistical evidence

Question	Test	Result
Q2 services	Trend test (prop.trend.test)	X-squared = 461.2, $p < 2.2e-16$
Q2 services	Chi-square, 0-1 vs 2+ services	X-squared = 301.4, $p < 2.2e-16$
Q3 payment	Chi-square payment method	X-squared = 648.1, $p < 2.2e-16$
Q3 billing	Chi-square paperless billing	X-squared = 258.3, $p < 2.2e-16$
Q4 senior	Chi-square	X-squared = 159.4, $p < 2.2e-16$
Q4 partner	Chi-square	X-squared = 158.7, $p < 2.2e-16$
Q4 dependents	Chi-square	X-squared = 433.7, $p < 2.2e-16$
Q4 gender	Chi-square	X-squared = 0.48, $p = 0.49$ (not significant)
Q5 DSL / fiber optic	Wilcoxon, charges by churn	$p < 2.2e-16$ (churners pay less)
Q5 no internet	Wilcoxon, charges by churn	$p = 0.004$ (difference only \$0.70)
Q6 cities	Chi-square, 8 cities	X-squared = 4.43, $p = 0.73$ (not significant)
Q7 CLTV	Wilcoxon, CLTV by churn	$p < 2.2e-16$

Question 1

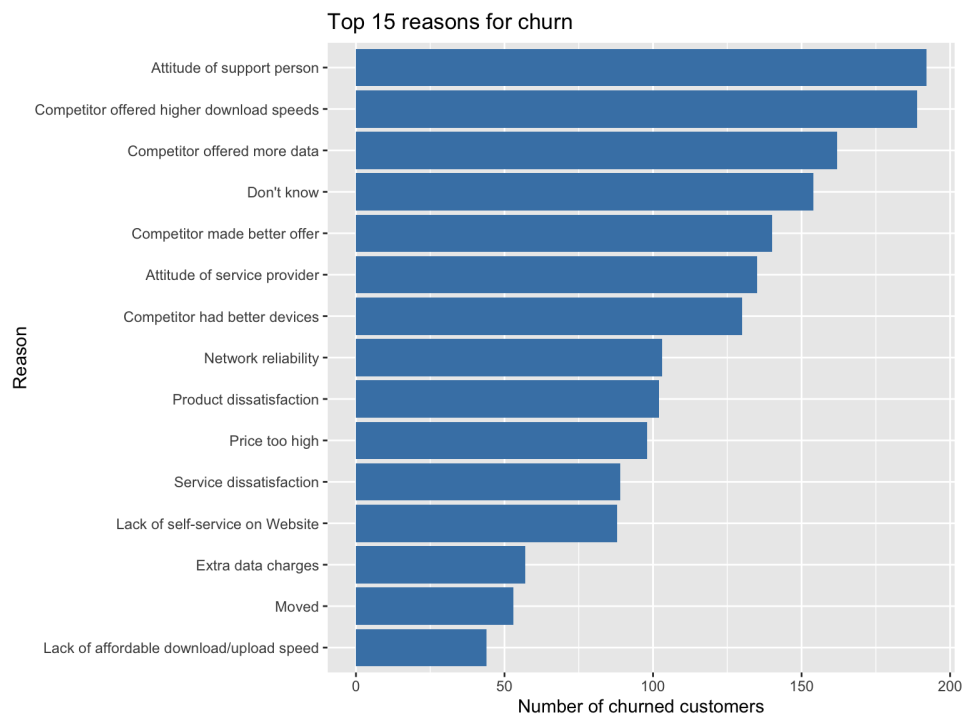
Q1. What reasons do churned customers give for leaving?

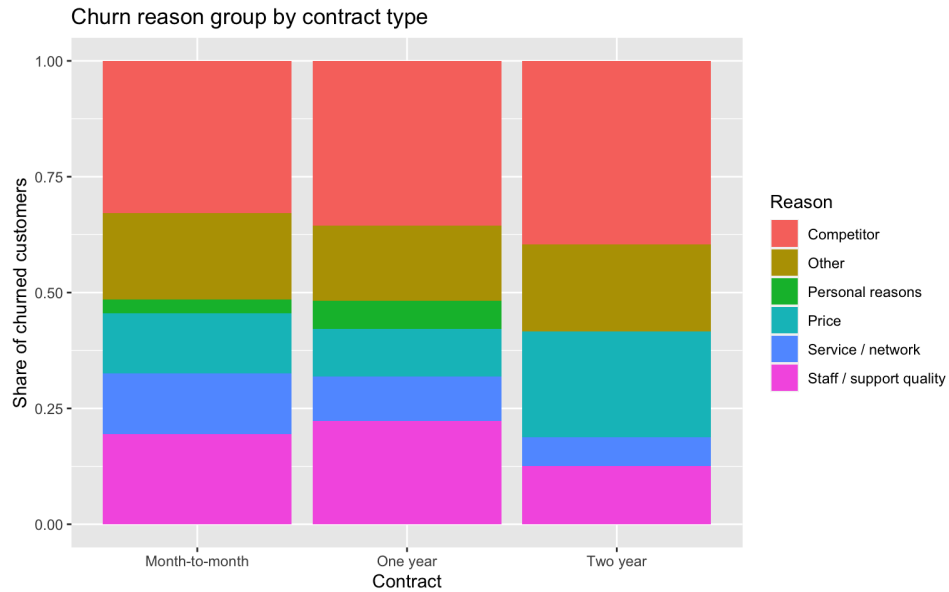
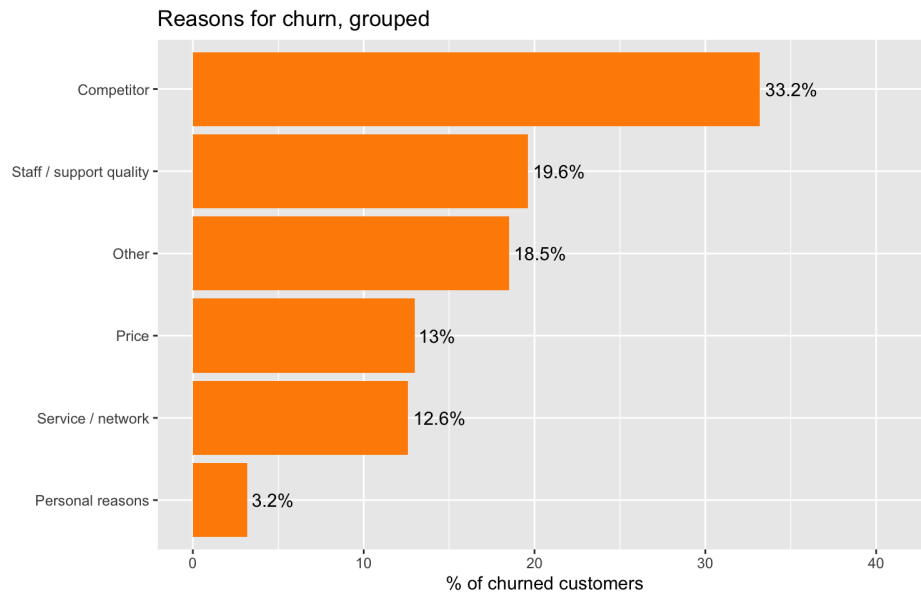
Answer: Mainly competitors (33.2%) and staff/support quality (19.6%). Personal reasons such as moving are only 3.2%.

Explanation

Of the 1,869 customers who churned, every one has a recorded reason (20 different ones). Grouping them shows that one third of churn is about a competitor offering better prices, devices, speeds or more data, and one fifth is about the attitude or lack of expertise of support staff. Price on its own is 13% and network or service issues 12.6%.

The reason mix is similar across contract types, but 89% of churners (1,655 customers) are on month-to-month contracts, so that is where these reasons play out. Both main reasons are controllable: better competitive offers, and better training and service quality.





Question 2

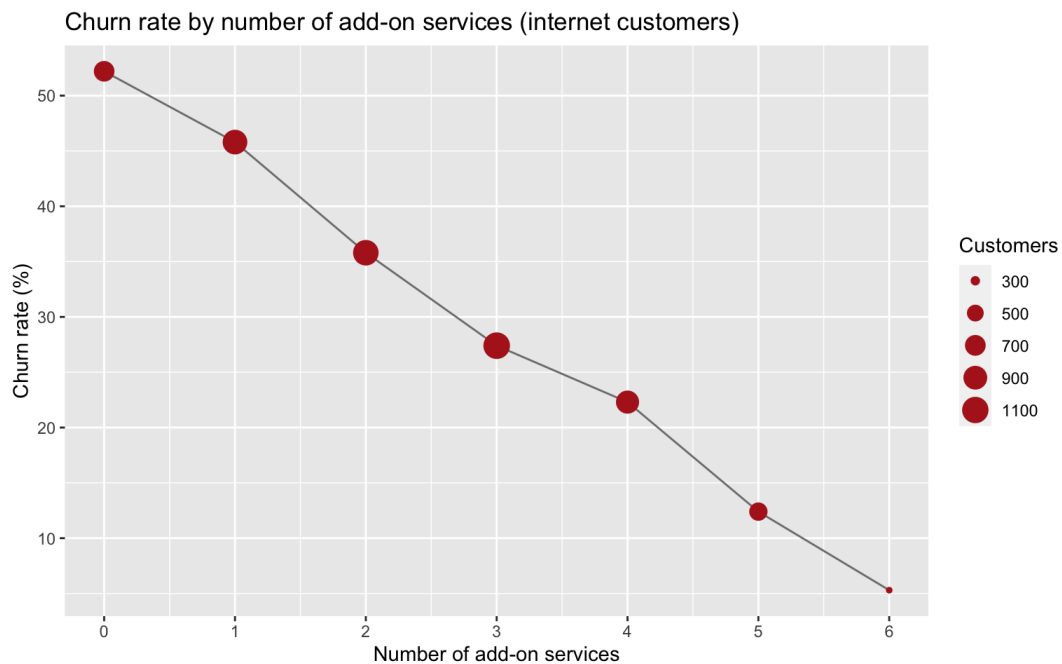
Q2. Do customers with more add-on services churn less?

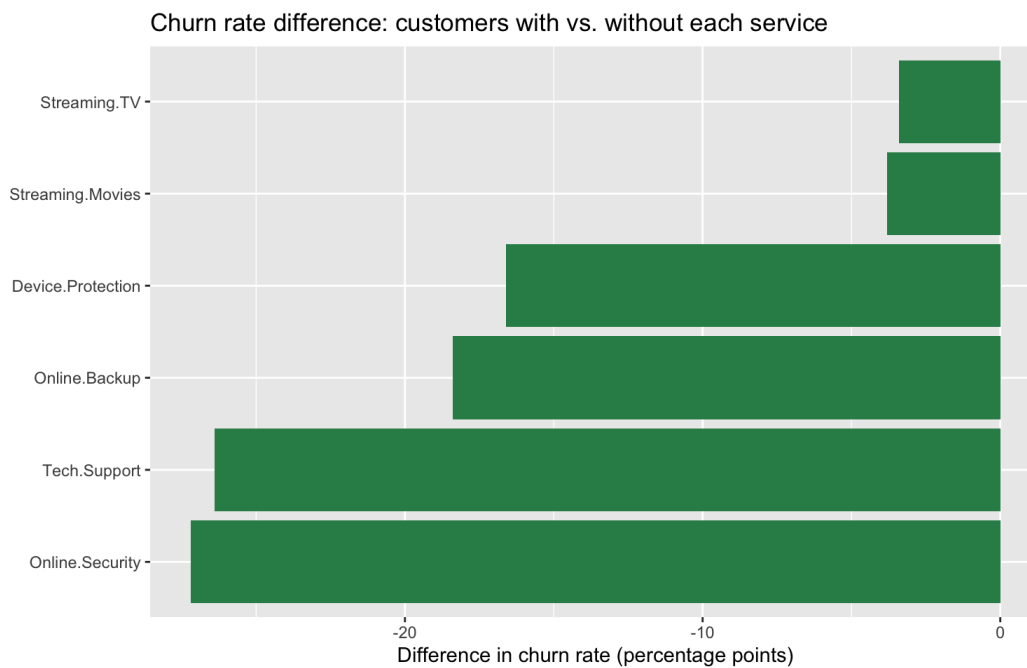
Answer: Yes. Churn falls steadily from 52% (no add-ons) to 5% (all six), but the type of service matters: Online Security and Tech Support are the most protective.

Explanation

Among internet customers, each extra service lowers churn (trend test $p < 2.2e-16$). Customers with 2 or more services churn much less than those with 0-1 (chi-square $p < 2.2e-16$). Looking at services one by one, Online Security (-27 points) and Tech Support (-26 points) show the biggest gaps, while the two streaming services show only -3 to -4 points.

Caution: this is an association. Customers who are already satisfied may simply buy more services.





Question 3

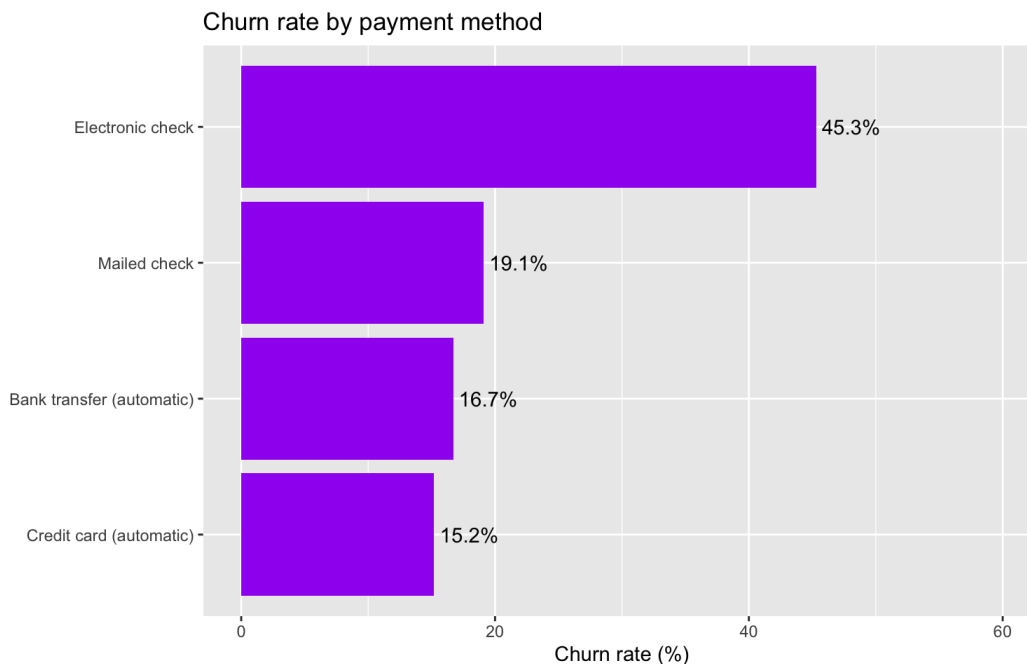
Q3. Are billing choices (payment method, paperless billing) linked to churn?

Answer: Yes. Electronic check users churn at 45.3% versus 15-19% for other methods, and the gap remains inside every contract type. Paperless billing is also linked to higher churn (33.6% vs. 16.3%).

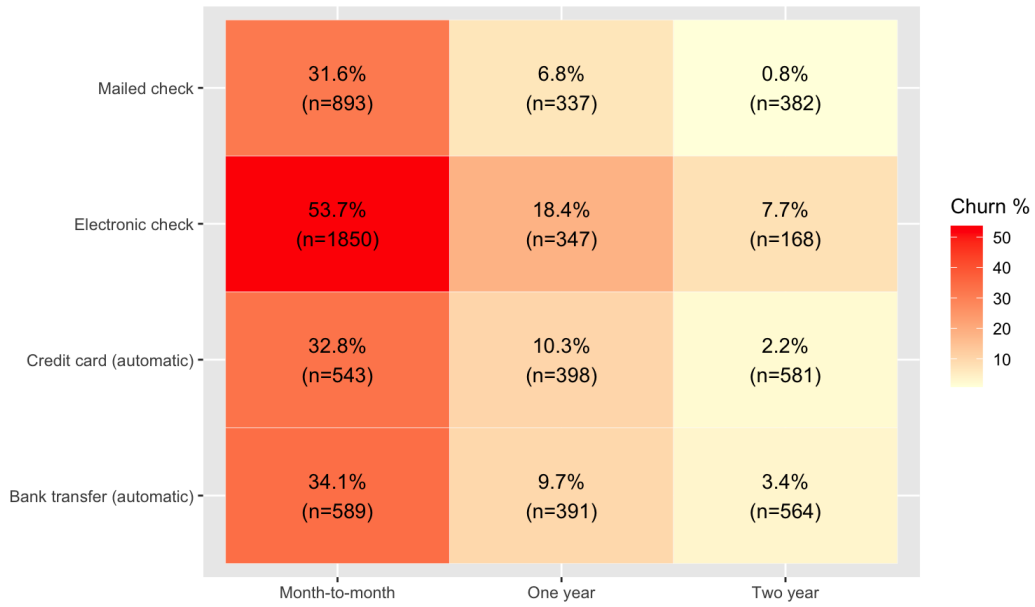
Explanation

Both chi-square tests are highly significant (payment $p < 2.2e-16$, paperless $p < 2.2e-16$). The heatmap shows that electronic check has the highest churn within month-to-month (53.7%), one-year (18.4%) and two-year (7.7%) contracts, so the effect is not only due to contract length.

Automatic payments (bank transfer, credit card) have the lowest churn. A possible reason is that automatic payment is a low-effort habit, so leaving takes an active decision.



Churn rate by payment method and contract



Question 4

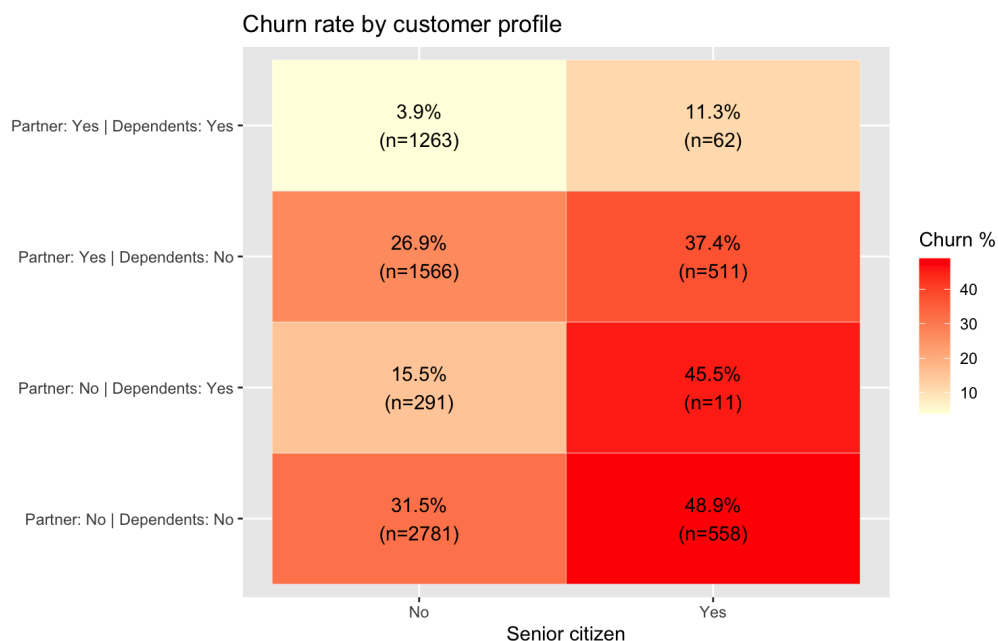
Q4. How do family situation and age relate to churn?

Answer: Seniors churn more (41.7% vs. 23.6%), customers with dependents churn far less (6.5% vs. 32.6%), and having a partner lowers churn (19.7% vs. 33.0%). Gender has no effect.

Explanation

The most at-risk profile is a senior with no partner and no dependents (48.9% churn, 558 customers). The safest is a non-senior with partner and dependents (3.9%, 1,263 customers). Chi-square tests are highly significant for senior, partner and dependents, and not significant for gender ($p = 0.49$).

In the combined model (Q8), dependents remain the strongest protective factor (odds ratio 0.24).



Question 5

Q5. Do churned customers pay more, and does it depend on the internet service type?

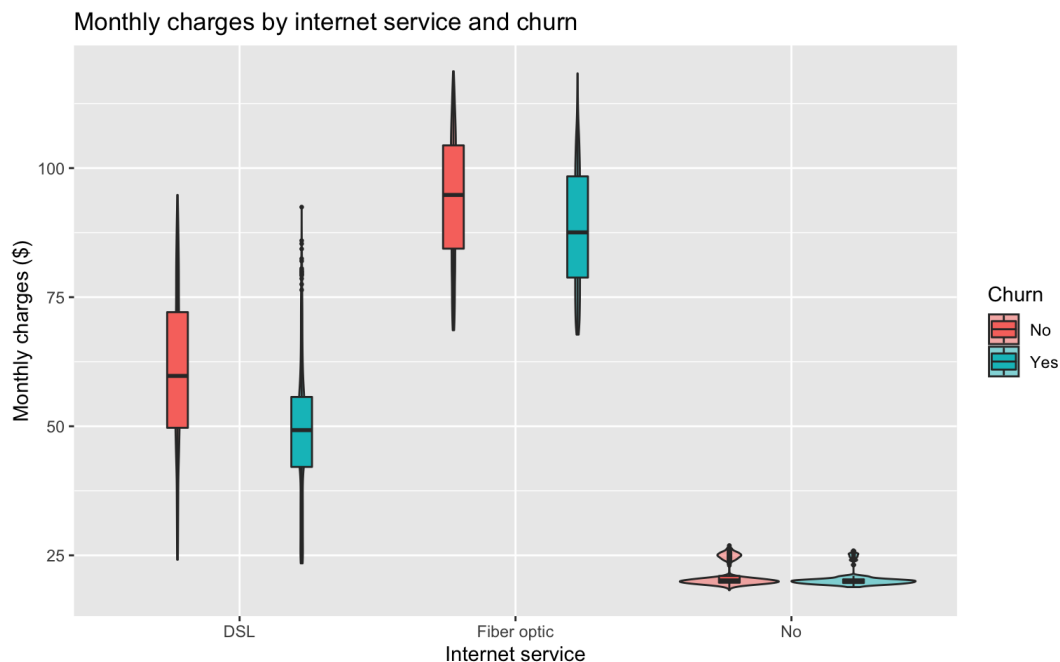
Answer: Overall yes (\$74 vs. \$61), but only because of the product mix. Inside each internet type, churners pay slightly less than those who stay.

Explanation

Fiber optic customers pay about \$90-95 per month and churn at 41.9%; DSL customers pay about \$55-60 and churn at 19.0%; customers without internet pay about \$20 and churn at 7.4%.

Because fiber customers are both expensive and likely to churn, the overall average for churners is higher. Within DSL, churners pay about \$11 less (mean \$49.1 vs. \$60.2); within fiber optic about \$6 less (\$88.1 vs. \$93.9). Wilcoxon tests are significant in all groups, but for customers without internet the gap is only \$0.70.

The conclusion is that the fiber optic product, not the price level itself, is associated with leaving. Competitor offers of higher download speed (Q1) may be part of the explanation.



Question 6

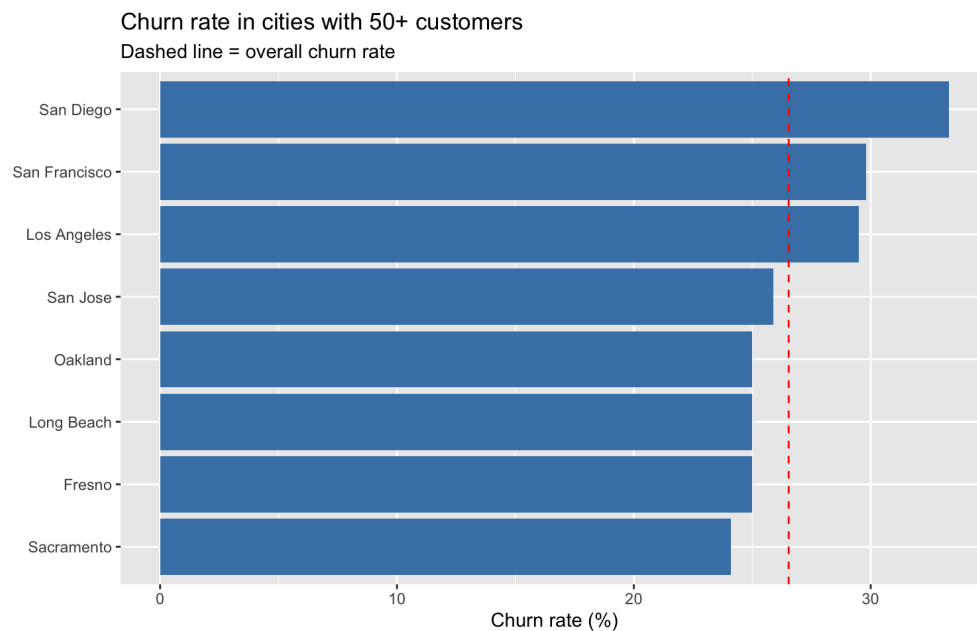
Q6. Does churn differ by location in California?

Answer: No meaningful difference. City churn rates range from 24% to 33%, but the differences are not statistically significant ($p = 0.73$).

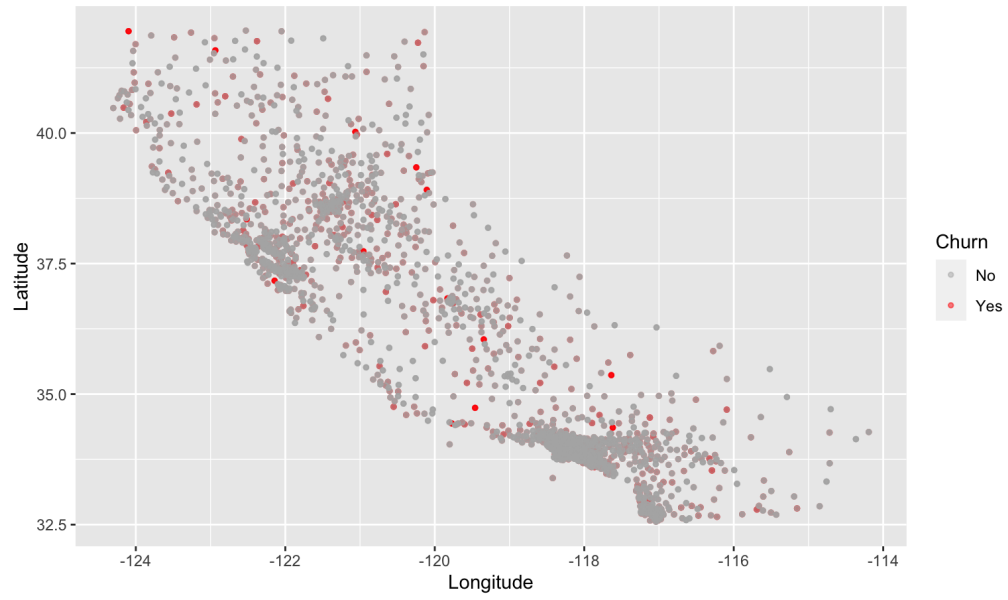
Explanation

We compared the eight cities with at least 50 customers. San Diego is highest at 33.3% and Sacramento lowest at 24.1%, around the overall 26.5%. The scatter map shows churned customers in every region of the state with no visible cluster.

Location is therefore not a main driver of churn compared with contract, services and billing.



Customers in California (churned in red)



Question 7

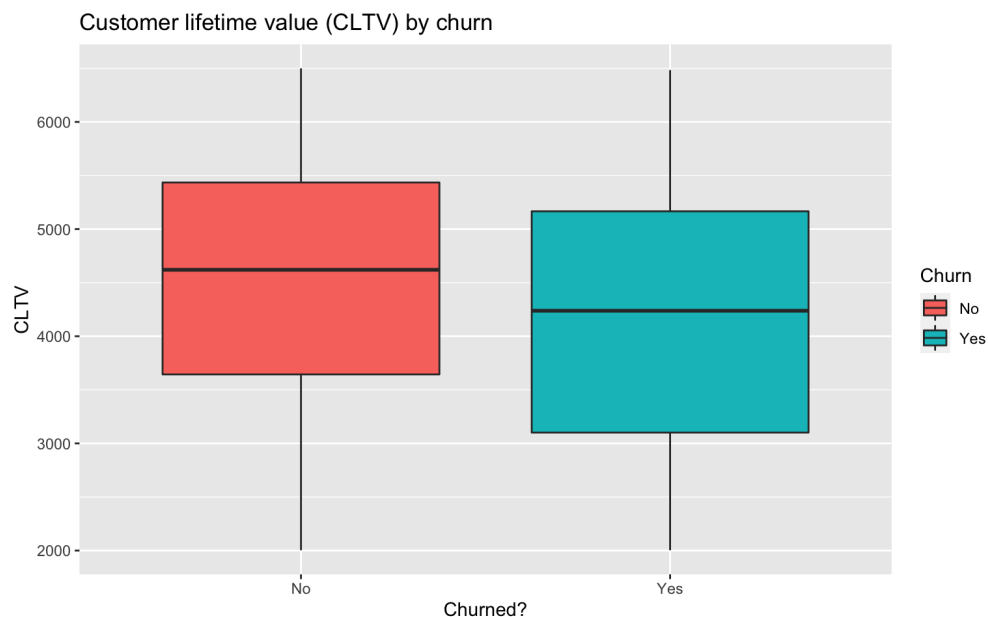
Q7. How much value is lost through churn?

Answer: Churned customers represent 30.5% of monthly revenue (\$139,131) while being 26.5% of customers, and 87% of the lost revenue is from month-to-month contracts.

Explanation

Churned customers pay more per month on average (\$74.4 vs. \$61.3) but have a lower lifetime value (mean CLTV \$4,149 vs. \$4,491, Wilcoxon $p < 2.2e-16$). The total CLTV of churned customers is about \$7.76 million. Of the lost monthly revenue, 86.9% comes from month-to-month, 10.1% from one-year and 3.0% from two-year contracts.

Note: IBM's *Churn Score* is strongly linked to the outcome (correlation 0.66). It was deliberately **not** used as a predictor, because it comes from IBM's own prediction and would cause data leakage.



Question 8

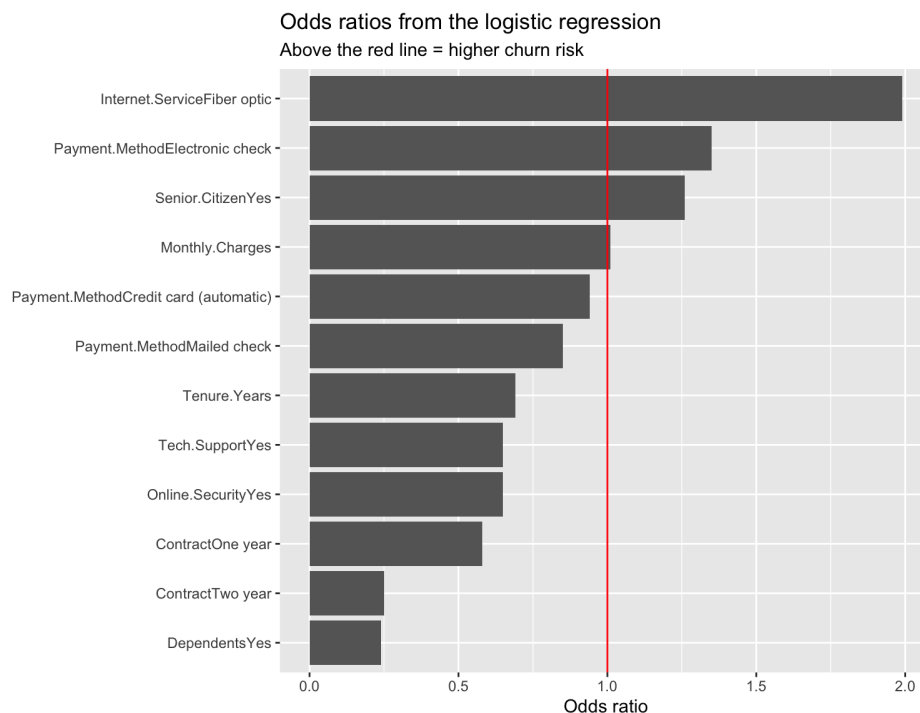
Q8. Which factors still matter when we look at them together?

Answer: Dependents, contract length, tenure, Online Security / Tech Support (all protective) and fiber optic internet, electronic check payment and senior age (all risky).

Explanation

A logistic regression on internet customers (70% training, 30% test, seed 123) gives odds ratios: fiber optic 1.99, electronic check 1.35, senior 1.26, tenure per year 0.69, Tech Support 0.65, Online Security 0.65, one-year contract 0.58, two-year contract 0.25 and dependents 0.24. Credit card and mailed check are not different from bank transfer, and monthly charges are only borderline significant ($p = 0.06$).

On the 30% of customers not used to fit the model, accuracy is about 77%, compared with 69% for always predicting 'No churn'. This is a useful check, not a final prediction model.



Summary of findings and implications

- **26.5%** of customers churned. Churn is mostly **preventable**: competitors (33%) and staff/support quality (20%) are the main reasons.
- **Protective services** (Online Security, Tech Support) go together with much lower churn (52% down to 5% as services increase).
- **Billing**: electronic check and paperless billing customers churn more, even within the same contract type.
- **Profile**: seniors living alone are highest risk; customers with dependents are safest. Gender does not matter.
- **Product, not price**: fiber optic customers churn at 41.9%. Within the same product, churners do not pay more.
- **Location** does not explain churn ($p = 0.73$).
- **Money**: churners are 30.5% of monthly revenue, 87% of it from month-to-month contracts.
- **Together**: dependents, contract length, tenure, fiber optic, electronic check and protective services are the key factors (77% test accuracy).

What the company could do

- Encourage month-to-month customers to move to longer contracts, especially in their first months.
- Promote Online Security and Tech Support, for example as a free trial for new customers.
- Review the fiber optic offer (price versus speed) and the quality of support staff.
- Offer incentives to switch from electronic check to automatic payment.
- Give special attention to senior customers who live alone.

Limitations and open questions

- The data is observational, so all results are associations, not proof of cause.
- Some groups are small (for example churners with long contracts and seniors with dependents but no partner).
- Churn reasons are self-reported; 154 customers answered 'Don't know'.
- Open questions: would early contact with tech support keep customers? Why does fiber optic churn more? Could a more flexible model (for example a decision tree) predict churn better?